Practical – 8_b

Develop a program that demonstrates copy-on-write behaviour after a fork(). Analyze memory usage before and after modifying data in the child process.

```
  GNU nano 8.7.1                                cow.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    int x = 100;

    if (fork() == 0)
    {
        printf("Child Before=%d\n", x);

        x = 500;

        printf("Child After=%d\n", x);
    }
    else
    {
        sleep(2);

        printf("Parent=%d\n", x);
    }

    return 0;
}
```

Output :

```
GNU bash — copy-on-write after fork()

saanvi_kompelli@DESKTOP-2U40RG7:~$ nano cow.c
saanvi_kompelli@DESKTOP-2U40RG7:~$ gcc cow.c -o cow
saanvi_kompelli@DESKTOP-2U40RG7:~$ ./cow
Child Before=100
Child After=500
Parent=100
Copy-on-write: child changes its private copy of data.
Parent value remains unchanged.
saanvi_kompelli@DESKTOP-2U40RG7:~$
```